Max Path Sum In Binary Tree

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The problem

- Maximum sum over any path (node-to-node, no repeats).

Intuition

- At each node, the best path either bends through it or continues upward through one child.

Approach

- 1. Post-order returns the best downward branch
- 2. A bending path is node plus both positive branches
- 3. Track the global max

Java solution

```java
public class MaxPathSumInBinaryTree {
    public static int maxPathSum(BinaryTree t) { int[] best = {Integer.MIN_VALUE}; branch(t, best); return best[0]; }

    private static int branch(BinaryTree t, int[] best) {
        if (t == null) return 0;
        int l = Math.max(0, branch(t.left, best)), r = Math.max(0, branch(t.right, best));
        best[0] = Math.max(best[0], t.value + l + r);
        return t.value + Math.max(l, r);
    }
}
```

Complexity

- Time: O(n)
- Space: O(h)

Recap

- At each node, the best path either bends through it or continues upward through one child.
- Complexity: O(n)